DAMO 699: Capstone Project

Poster Presentation Guidelines

The capstone poster presentation is the final stage of your applied analytics project. It is designed to simulate a professional research or industry showcase, where you communicate complex analytical work clearly, concisely, and visually.

Your poster should demonstrate not only technical competence, but also clarity of reasoning, analytical rigor, and professional communication.

1. Poster Size & Format Requirements

- Recommended Size:
  - 36 inches (height) × 48 inches (width)
  - Landscape orientation is strongly recommended.
- Alternative Acceptable Size (if printing constraints apply):
  - 24 inches × 36 inches (minimum acceptable size)
- Format:
  - Printed, full-color poster or black and white poster
  - Professional layout (PowerPoint, Canva, Adobe Illustrator, or similar software)
  - Fonts must be readable from at least 3–4 feet away

2. Required Poster Sections

Your poster must include the following clearly labeled sections:

1. Title, including: • Project title • Your name(s) • Course name	**2. Research Problem / objective of your capstone project** • This will vary depending on your project, please consult with your supervisor if unsure.

• Instructor name • Institution (students are free to use the UNF logo) • Date	
3. Research Question(s) and/or Hypotheses	**4. Data Description or preprocessing performed**
5. Analytical approach used (e.g., regression, classification, clustering, forecasting, etc.) • Justification for chosen method • Model diagnostics (if applicable)	**6. Key Findings** • Major statistical results • Significant relationships • Model performance metrics • Clear interpretation (not just output screenshots)

7. Visualizations
• Graphs must be:
  ○ Clean
  ○ Properly labeled
  ○ Interpreted
• Every chart must support a claim.
	8. Strategic / Practical Implications
	• What do the findings mean?
	• What should decision-makers do?

9. Limitations & Future Research • Data limitations • Model limitations • Suggestions for future improvements	

Avoid overcrowding with raw output. Present summarized results visually.

3. Design Expectations

Your poster should:

- Use consistent formatting
- Avoid dense paragraphs
- Use bullet points for clarity
- Maintain visual balance (text vs visuals)
- Use high contrast between background and text
- Avoid excessive decorative elements
- Think professional conference standard.

4. Presentation Expectations

- Each group will stand beside their poster.
- You must be prepared to:
    - Explain your project clearly in 5–7 minutes
    - Answer questions
    - Defend your choices
    - Interpret results without reading directly from the poster
- Professional attire is expected.

5. Setup Requirement (Mandatory)

All students must:

- Arrive at least 30 minutes early on the day of the presentation
- Use this time to:

- Set up your poster
- Ensure proper placement
- Test any additional materials (if applicable)
- Late arrival may impact your presentation grade.

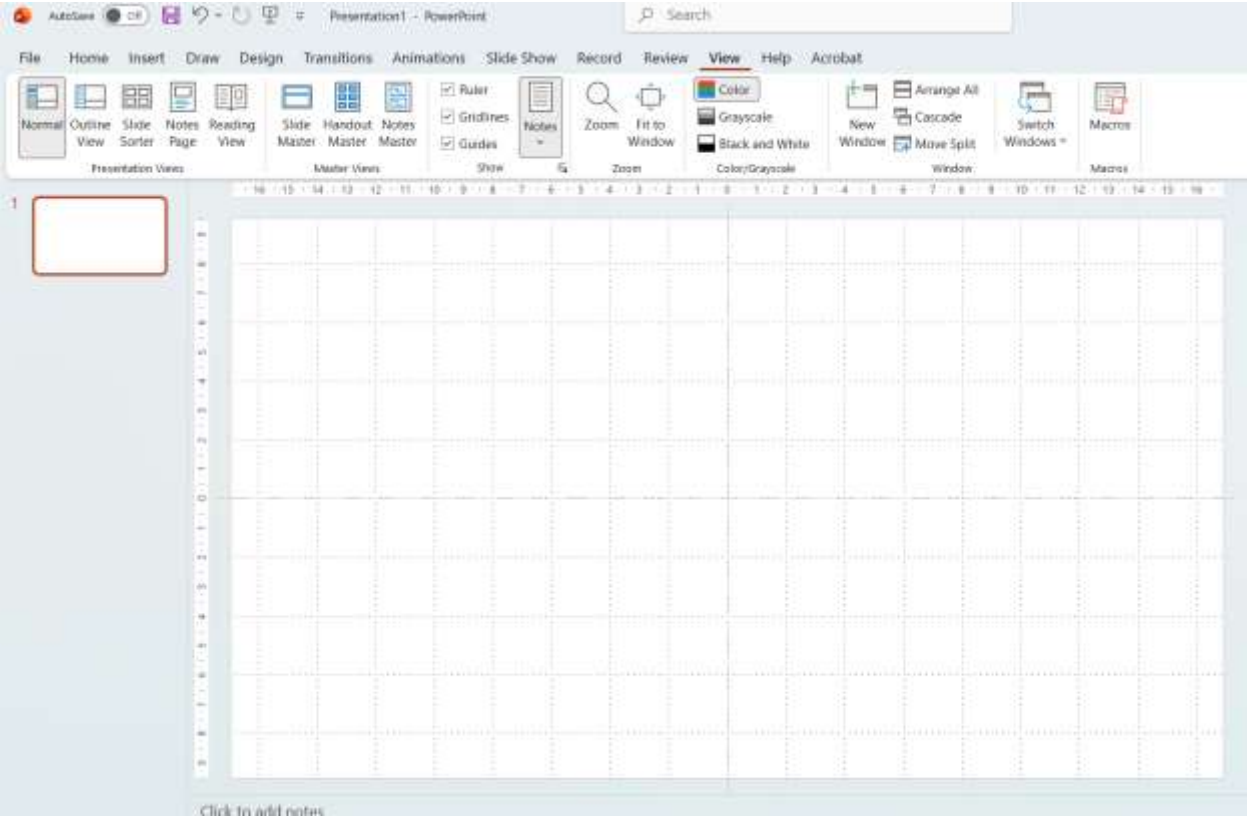
6. Assessment Criteria (High-Level Overview)

Your poster presentation will be evaluated based on:

- Analytical rigor
- Clarity of research design
- Proper use of data
- Interpretation accuracy
- Visual communication quality
- Professionalism during Q&A
- Overall coherence of the project

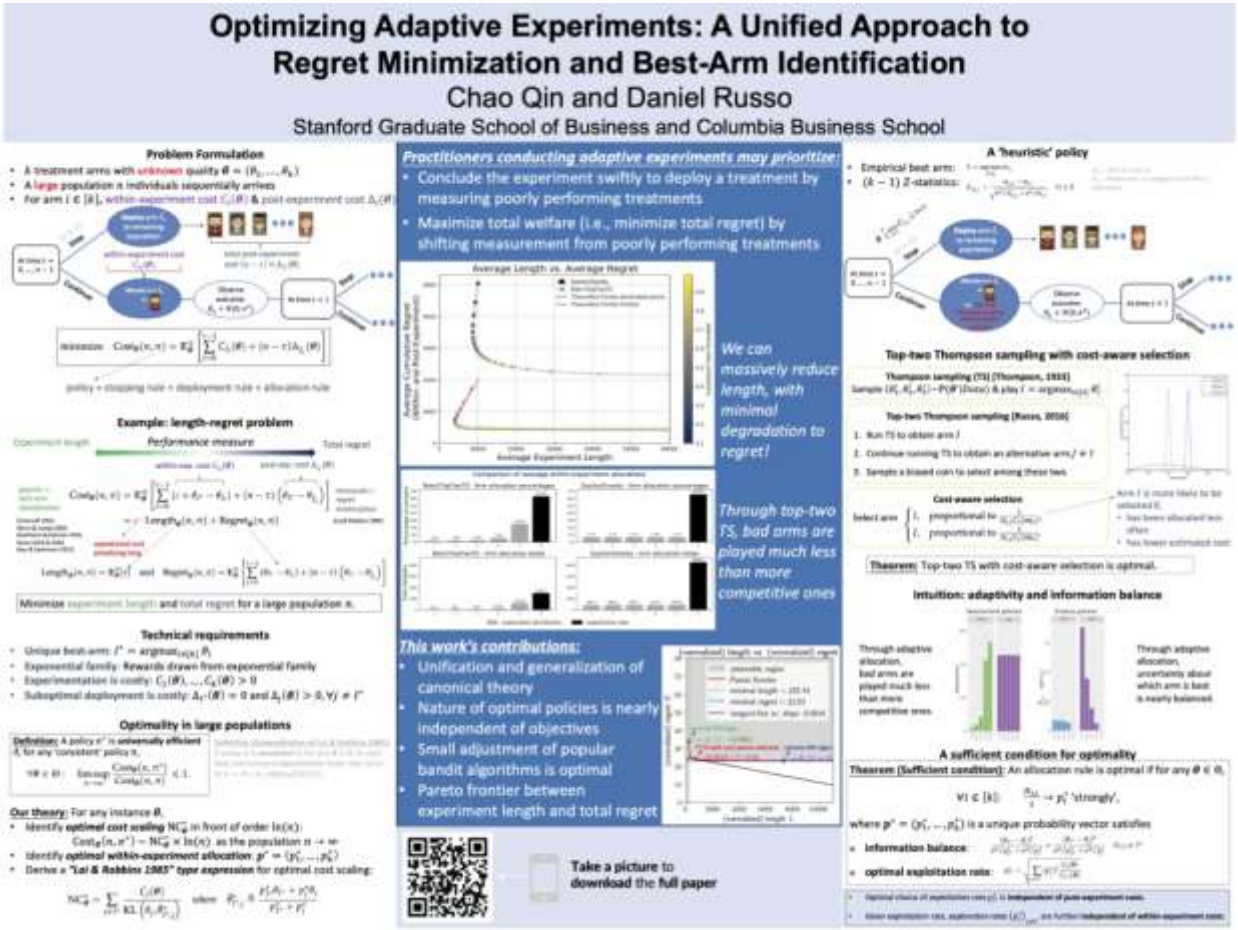
Poster Printing: TPH The Printing House **Location:** 6700 Morrison St Unit 3, Niagara Falls, Canada (walking distance from campus) **Price:** $25 +tax for coloured poster **Format:** Please save your file as a PDF **IMPORTANT:** Must bring your poster to the printer no later than Wednesday March 18 in order to pick up the poster by Friday. Please note that they are not open on the weekend, so students presenting in the morning, you will need to pick up your poster by Friday. *Please note you are not required to use this location, but can use another location of your choice*

If you are using PowerPoint for the presentation:



Template source and guide: https://www.posternerd.com/sciposters-templates?srsltid=AfmBOor6Xvxv7SKLKTW-3opQb8FbudpO7pY-39bUv0IkP2ALOUo5dZdp

Samples:


Optimizing Adaptive Experiments: A Unified Approach to Regret Minimization and Best-Arm Identification
Chao Qin and Daniel Russo
Stanford Graduate School of Business and Columbia Business School
Problem Formulation
Example: length-regret problem
Technical requirements
Optimality in large populations
Our theory
Practitioners conducting adaptive experiments may prioritize:
Conclude the experiment swiftly to deploy a treatment by measuring poorly performing treatments
Maximize total welfare (i.e., minimize total regret) by shifting measurement from poorly performing treatments
Average Length vs. Average Regret
We can massively reduce length, with minimal degradation to regret!
Through top-two TS, bad arms are played much less than more competitive ones
This work's contributions:
Unification and generalization of canonical theory
Nature of optimal policies is nearly independent of objectives
Small adjustment of popular bandit algorithms is optimal
Pareto frontier between experiment length and total regret
Take a picture to download the full paper
A 'heuristic' policy
Top-two Thompson sampling with cost-aware selection
Cost-aware selection
Theorem: Top-two TS with cost-aware selection is optimal.
Intuition: adaptivity and information balance
A sufficient condition for optimality

Why Did the Wildlife Cross the Road?

Effects of Season, Precipitation, and Temperature on Road Mortality on the 1000 Islands Parkway

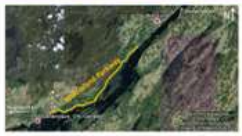
Lyn Garrah, Queen's University MES candidate

Parks Canada Parcs Canada

INTRODUCTION

- Road mortality may lead to isolated or extirpated wildlife populations
- Seven of the 14 snake and turtle species in the study area are COSEWIC listed species at risk
- Identifying road mortality patterns and the reasons for them can help develop effective management strategies

STUDY LOCATION



- The 1000 Islands Parkway is a 37 km two lane road
- Located immediately adjacent to the St. Lawrence River between Gananoque and Brockville, Ontario

METHODS

- Cycled the Parkway 3-4 times weekly from June 2nd to September 30th, 2010 (16 weeks)
- Recorded the location of vertebrates on the road using a hand-held GPS
- Road kill was identified by class or species

DISCUSSION

- **Frogs** cross roads primarily based on season (in July and early August) with increased movement in warm, wet weather
- **Snakes** are killed on roads in the fall, likely when returning to hibernacula or basking on roads on warmer days in this cooler season
- **Turtles** are killed in June when females seek nesting sites, and more likely when it rains
- **Birds** are killed primarily during migration and breeding season, generally when it is not raining
- **Mammals,** being endotherms, are less likely to be influenced by weather variables

FUTURE WORK

- Analyze 2008 and 2009 road kill data relationship with climate, season and traffic patterns
- Analyze reptile road kill age and sex from 2010 data
- Evaluate effectiveness of experimental fencing adjacent to culverts in mitigating road kill
- Evaluate culvert use by analyzing motion sensor digital camera images

ACKNOWLEDGEMENTS

- Parks Canada for financial and logistic support
- This study builds on earlier research on the same topic conducted by the Geomatics and Landscape Ecology Research Laboratory at Carleton University
- NSERC Discovery Grant funding to supervisor R. Danby

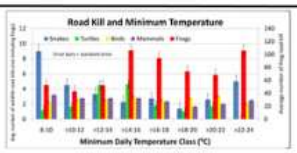
PRELIMINARY RESULTS

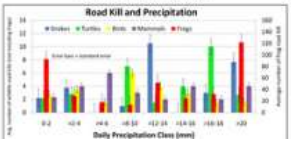
Frogs (n=4170)

- 50.5% found over 5 weeks (July 9 to August 16) of 16 week study
- Greater frog road kill when temperature >14 °C
- Greater frog road kill when higher precipitation

Snakes (n=122)

- 33.6% found in the last week of September
- Greater snake kills when temperature >18 ° C
- Greater snake kills when higher precipitation





Turtles (n=82)

- 58.6% found in June
- Greater turtle kills when higher precipitation

Birds (n= 128)

- 84.4% found in June and July
- Less bird road kill with increasing precipitation

Mammals (n=126)

- No season, precipitation or temperature relationship